ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST 1 - JANUARY-2026

CET05103 - BASIC VISUAL PROGRAMMING

SEMESTER: III **YEAR: 2**

DURATION: 1 Hr 30 Min. **MAX. MARKS: 50**

CURATED ANSWERS PROVIDED BY 4LAZIE - SMART IN BRAIN

PART A: SHORT ANSWER QUESTIONS

Q1. What does the acronym BASIC stand for?

Answer: BASIC stands for Beginner's All-purpose Symbolic Instruction Code.

Smart Tip: *Remember "Beginner's" as it was originally designed for students to easily learn programming.*

Q2. What is meant by "Visual" in Visual Basic?

Answer: "Visual" refers to the method used to create the Graphical User Interface (GUI). Instead of writing lines of code to build an interface, you can simply drag and drop pre-built objects (like buttons and textboxes) onto a screen.

Q3. What is a Module in Visual Basic?

Answer: A Module is a file (with a .vb extension) that contains code such as variables, constants, and procedures which can be accessed from anywhere within the project. It helps in organizing code and making it reusable.

Q4. List two numeric data types supported in Visual Basic.

Answer:

- **Integer:** Used for whole numbers (e.g., 10, -5).
- **Double:** Used for numbers with decimals (e.g., 3.14, -0.5).

Q5. What is the purpose of "Exit For"?

Answer: "Exit For" is a statement used to immediately terminate a For or For Each loop before it has finished all its iterations. It transfers execution to the statement immediately following the loop.

Q6. State the keyword used to declare a variable in Visual Basic.

Answer: The keyword 'Dim' (short for Dimension) is used to declare a variable in Visual Basic.

Q7. What is the purpose of Console.WriteLine()?

Answer: Console.WriteLine() is a built-in method used to display output text or data on the console screen, followed by moving the cursor to a new line.

PART B: EXPLANATORY QUESTIONS

Q8. Explain why visual basic has slower execution speed?

Answer: Visual Basic typically has a slower execution speed because it is an interpreted, managed language (compiled to Intermediate Language, then interpreted by the runtime) rather than being compiled directly to native machine code. Its heavy reliance on GUI components and runtime libraries also adds overhead.

Q9. Explain the difference between method and function.

Answer:

- **Function:** A block of code that performs a specific task and ALWAYS returns a value to the calling code.
- **Method (Sub procedure):** A block of code that performs a specific task but DOES NOT return a value. In VB, these are called 'Sub' procedures.

Q10. Explain how to declare and initialize an array in visual basic with examples.

Answer: An array is declared using the 'Dim' keyword, followed by the variable name, parentheses () to indicate the size, and the 'As' keyword for the data type.

```
' Declaration
Dim numbers(4) As Integer

' Initialization
numbers(0) = 10
numbers(1) = 20
```

Smart Tip: *You can also declare and initialize on one line: Dim numbers() As Integer = {10, 20, 30}*

Q11. Discuss the limitations of Visual Basic.

Answer:

- **Platform Dependency:** It is primarily designed to run on Windows operating systems.
- **Slower Execution:** As a managed language, it is slower compared to low-level languages like C or C++.
- **Memory Usage:** It consumes more memory due to the overhead of the .NET Framework and garbage collection.

Q12. Provide a short history of Visual Basic.

Answer: Visual Basic was created by Microsoft and first released in 1991 as a modern, event-driven successor to the older BASIC language. It revolutionized programming by introducing a drag-and-drop graphical interface builder. In 2002, Microsoft released Visual Basic .NET (VB.NET), integrating it fully into the .NET framework and making it a true object-oriented language.

Q13. Write a simple Visual Basic program to enter and display user name and age.

Answer: Here is a simple console application to achieve this:

```
Module Module1
    Sub Main()
        Console.Write("Enter your name: ")
        Dim name As String = Console.ReadLine()
```

```
        Console.Write("Enter your age: ")
        Dim age As Integer = Convert.ToInt32(Console.ReadLine())

        Console.WriteLine("Hello " & name & ", you are " & age & " years old.")
        Console.ReadLine()
    End Sub
End Module
```

PART C: EXPLANATORY AND PROGRAMMING QUESTIONS

Q14. Analyze the advantages and disadvantages of using the Visual Basic IDE for system-level versus application-level programming.

Answer:

• **Application-Level Programming (e.g., Desktop Apps, Inventory Systems):**

- Advantages: VB is excellent here. The IDE provides Rapid Application Development (RAD) tools, meaning you can build GUIs by dragging and dropping buttons. It has rich built-in libraries for database connections and web services, speeding up development.
- Disadvantages: The final application requires the end-user to have the .NET framework installed on their machine.

• **System-Level Programming (e.g., Operating Systems, Device Drivers):**

- Advantages: Code is very readable and safe from common memory leaks due to the Garbage Collector.
- Disadvantages: VB is highly discouraged for system-level tasks. It lacks direct, low-level memory access (like pointers in C/C++) which is crucial for interacting directly with hardware. Furthermore, the intermediate compilation adds execution overhead which is unacceptable for high-performance system drivers.

Q15. Discuss the various Decision-Making structures available in Visual Basic.

Answer:

Decision-making structures allow a program to take different paths depending on certain conditions. The main structures are:

1. If...Then...Else: Used to evaluate a condition. If true, it runs one block of code; if false, it runs another.

```
If score >= 50 Then
    Console.WriteLine("Pass")
Else
    Console.WriteLine("Fail")
End If
```

2. If...ElseIf...Else: Used when you need to test multiple consecutive conditions.

3. Select Case: Tests a single variable against a list of matching values. It is cleaner and faster than using multiple ElseIf statements.

```
Dim grade As String = "A"
Select Case grade
    Case "A"
        Console.WriteLine("Excellent")
    Case "B"
        Console.WriteLine("Good")
    Case Else
        Console.WriteLine("Invalid")
End Select
```

Q16. Analyze the different types of Loops in Visual Basic with simple example program.

Answer:

Loops are used to execute a block of code repeatedly. The main types are:

1. For...Next Loop: Best when you know exactly how many times you want to iterate.

```
For i As Integer = 1 To 3
    Console.WriteLine("Iteration " & i)
Next
```

2. While...End While Loop: Continues to loop as long as the specified condition evaluates to True.

```
Dim count As Integer = 1
While count <= 3
    Console.WriteLine("Count is " & count)
    count += 1
End While
```

3. Do...Loop: Similar to While, but can guarantee the code runs at least once if you check the condition at the end (Do...Loop Until).

Q17. Explain in details about multi dimension array and the looping process with an example program.

Answer: A multi-dimensional array organizes data in multiple dimensions, most commonly a 2D array which acts like a table with rows and columns. Think of it as a spreadsheet. To read or print data from a 2D array, we use 'Nested Loops'. The outer loop typically controls the rows, and the inner loop controls the columns.

Example Scenario: Storing scores of 2 students in 3 subjects.

```
Module Module1
    Sub Main()
        ' 2D array: 2 rows (students) and 3 columns (subjects)
        Dim scores(,) As Integer = { {85, 90, 78}, {88, 92, 80} }

        ' Outer loop for rows (Students 0 to 1)
        For row As Integer = 0 To 1
            Console.Write("Student " & (row + 1) & " scores: ")

            ' Inner loop for columns (Subjects 0 to 2)
            For col As Integer = 0 To 2
                Console.Write(scores(row, col) & " ")
            Next

            Console.WriteLine() ' Move to next line for the next student
        Next
        Console.ReadLine()
    End Sub
End Module
```